Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
    (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

*(i-1+arr) = i%3

0 1 2 0

Output
0 1 2 0
1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Yes, temp is a local variable of func2. After a = &temp; a starts pointing to temp & in the actual array there is no change. Once func2 is over a & temp get released from memory ~~Original~~ array is same as before

Output is
0120
0120

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

Output: 320 160

sizeof(long int) = 8 & sizeof(int) = 4

a1 = arr2 + 50 * 8 as arr2 is long type
a2 = arr2 + 10 * 8
⇒ a1 - a2 = 40 * 8 = 320

b1 = arr1 + 50 * 4 as arr1 is int type
b2 = arr1 + 10 * 4
b1 - b2 = 40 * 4 = 160